

Auditing Deep Learning for Cardiac MRI Through Semantic Transitions and Conflict-Aware Production Rules

Cardiac magnetic resonance (CMR): measurable semantics and guarded rules

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Prediction quality leaves an audit gap

Table 1. Accuracy and absolute error

Metric	Khened	Isensee	Study
Diagnostic accuracy	0.960	0.920	0.860
LVEDV MAE (mL)	4.20	5.10	4.88
LVEF MAE (pp)	2.50	2.10	9.17
RVEF MAE (pp)	5.30	4.70	7.52
Mass MAE (g)	6.30	7.30	3.83
Rule coverage	—	—	0.640
Rule balanced acc.	—	—	0.644

The audit gap

Link recognizable measurements to rules, provenance, conflict, and safe abstention.

Table 2. Correlation, bias, and audit outputs

Index	Khened	Isensee	Study
LVEDV r	0.997	0.997	0.710
LVEDV bias (mL)	+0.6	+1.5	−2.12
LVEF r	0.989	0.991	0.863
LVEF bias (pp)	−0.5	+0.5	−1.93
RVEF r	0.858	0.925	0.815
RVEF bias (pp)	−2.2	−3.0	+3.56
Mass r	0.990	0.986	0.517
Mass bias (g)	−2.9	−4.0	−1.46
Coverage	—	—	0.640
Abstention	—	—	0.360
Conflict	—	—	0.313

Table 2: Pearson r and signed bias. pp = percentage points. Study: transition + semantic baseline / rule audit.

Table highlights: **blue**: reported result; **amber**: guarded abstention; **rose**: error / drift flag; **green**: proxy evidence.

Different splits and objectives: context, not a controlled ranking.

O. Bernard et al. “Deep Learning Techniques for Automatic MRI Cardiac Multi-Structures Segmentation and Diagnosis: Is the Problem Solved?” *IEEE Transactions on Medical Imaging*, 37(11):2514–2525, 2018. doi:10.1109/TMI.2018.2837502.

Make every audit output traceable

Purpose: translate fixed CMR representations into measurable semantics and guarded rules.

01 / Align

Patient identity, feature QC, and source provenance.

02 / Reconstruct

128 formal coordinates → 28 CMR measurements.

03 / Audit

Rule support, conflict, missing evidence, and abstention.

ACDC = Automated Cardiac Diagnosis Challenge.

M&Ms = Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation.

Table 3. Cohorts and release-level quality control

Audit item	ACDC	M&Ms	Scope / interpretation
Patient-level cases	150	100	ACDC: 85 / 15 / 50 train / validation / test; M&Ms: external only.
Semantic validation checks	10/10	10/10	All programmed gates passed.
Native overlay panels	53	65	End-diastolic / end-systolic image-mask panels.
Semantic targets per case	28	28	Same semantic schema in both cohorts.
Pending manual review	—	—	92 cases across both cohorts; no dataset-specific split reported.

QC (quality control): 92 pending reviews across both cohorts; not confirmed errors or completed expert validation.

A frozen, patient-level audit chain

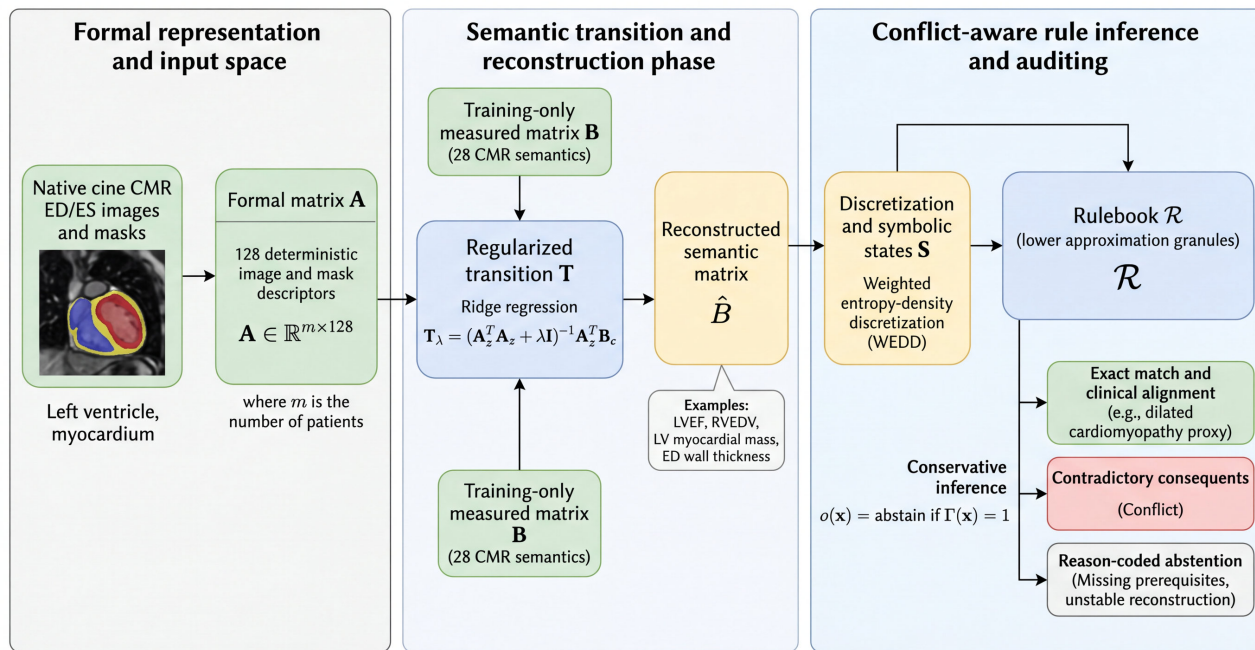


Figure 1. Framework architecture; the evaluated release substitutes training tertiles for the weighted entropy-density discretization (WEDD) block.

Algorithm 1

1. Lock cohorts + QC
2. Fit transition
3. Discretize
4. Induce rules
5. Guard inference
6. Align predicates
7. Evaluate fidelity
8. Test perturbations

Evaluated release

57 descriptors
projected to 128
coordinates.

The WEDD block is
approximated by
frozen training tertiles.

Ridge fit and frozen reconstruction

$$\mathbf{T}_\lambda = \arg \min_{\mathbf{T}} \|\mathbf{A}_z \mathbf{T} - \mathbf{B}_c\|_F^2 + \lambda \|\mathbf{T}\|_F^2. \quad (1)$$

$$\mathbf{T}_\lambda = (\mathbf{A}_z^\top \mathbf{A}_z + \lambda \mathbf{I})^{-1} \mathbf{A}_z^\top \mathbf{B}_c. \quad (2)$$

$$\hat{\mathbf{b}}(\mathbf{a}) = \mathbf{a}_z \mathbf{T}_\lambda + \boldsymbol{\mu}_B^\top. \quad (3)$$

Training-only preprocessing

Median imputation; 1st–99th percentile clipping; formal standardization; target centering. Five-fold training cross-validation selected $\lambda = 3$.

$\mathbf{A}_z$: standardized formal features; $\mathbf{B}_c$: centered targets; $\boldsymbol{\mu}_B$: training means. The 28 targets cover volumes, EF, mass, wall thickness, and shape.

Evaluated representation

57 deterministic descriptors $\rightarrow$ 128 coordinates by a fixed Gaussian projection; 32 direct channels retained. Seed: 20260611.

Frozen tertile states

$$s_j(b) = \begin{cases} \text{low,} & b \leq q_{j,1/3}, \\ \text{mid,} & q_{j,1/3} < b \leq q_{j,2/3}, \\ \text{high,} & b > q_{j,2/3}. \end{cases} \quad (4)$$

This experiment uses deterministic descriptors and tertiles; transition coefficients express associations.

Compact production rules

$$r : \bigwedge_{j \in R_r} [s_j = v_j] \implies d = y_r. \quad (5)$$

At most 3 antecedents; support at least 3 cases; confidence at least 0.60.

Guarded inference

$$o(x) = \begin{cases} \text{abstain}[\rho(x)], & \Gamma(x) = 1, \\ y, & \Gamma(x) = 0 \text{ and } \mathcal{Y}(x) = \{y\} \end{cases}. \quad (6)$$

$\Gamma(x)$: guard; $\rho(x)$: reason code; $\mathcal{Y}(x)$: fired consequents. Guards cover severe QC, unstable reconstruction, missing prerequisites, no match, and contradictory consequents.

Physician-style alignment

$$\text{Sim}(P_y, R_i) = \max \left\{ 0, \frac{\sum_j w_j \phi_j(P_y, R_i)}{\sum_j w_j} - \gamma_c C_i - \gamma_m M_i - \gamma_u U_i + \gamma_a A_i \right\}. \quad (7)$$

ϕ_j : compatibility; w_j : clinical weights. C_i, M_i, U_i, A_i : conflict, missing evidence, instability, and safe abstention.

Eligibility before similarity

Only rules passing support, confidence, prerequisite, and validity gates are eligible.

Soft similarity never overrides the abstention guard. Cine-supported proxies remain evidence-bounded.

Table 4. Aggregate reconstruction

Split	SD-NRMSE	RMSE	MAE
ACDC train	0.510 ± 0.148	33.18	26.49
ACDC val.	0.814 ± 0.409	58.42	46.49
ACDC test	0.767 ± 0.246	51.65	38.64
M&Ms ext.	1.726 ± 1.177	85.35	69.81

$$\text{NRMSE}_j = \text{RMSE}_j / s_{j,\text{tr}}. \quad (8)$$

NRMSE: normalized root mean square error (training SD); SD: standard deviation. MAE: mean absolute error.

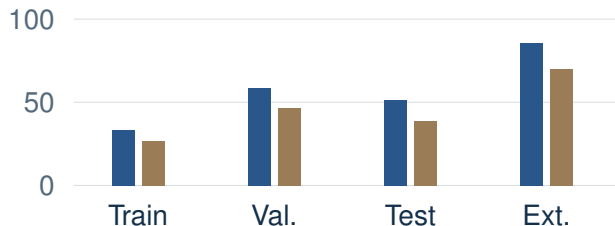


Figure 2. RMSE / MAE. M&Ms external bias: LVEDV −136.0 mL; mass −96.6 g.

External NRMSE is 2.25 times the held-out test error. Mixed-unit RMSE / MAE means are descriptive.

Table 5. Held-out ACDC target fidelity

Target	RMSE	MAE (95% CI)	r	Bias
Global ED wall thickness (mm)	0.83	0.59 (0.46–0.79)	0.928	−0.11
LVEF (pp)	11.38	9.17 (7.52–11.16)	0.863	−1.93
RVEF (pp)	8.87	7.52 (6.33–8.91)	0.815	+3.56
LVEDV (mL)	5.86	4.88 (4.02–5.75)	0.710	−2.12
RVEDV (mL)	9.75	6.37 (4.89–9.19)	0.766	−3.13
LV mass, ED (g)	4.60	3.83 (3.17–4.58)	0.517	−1.46

$n = 50$; 1,000-resample bias-corrected and accelerated (BCa) 95% confidence intervals (CIs). ED: end-diastolic; pp: percentage points.

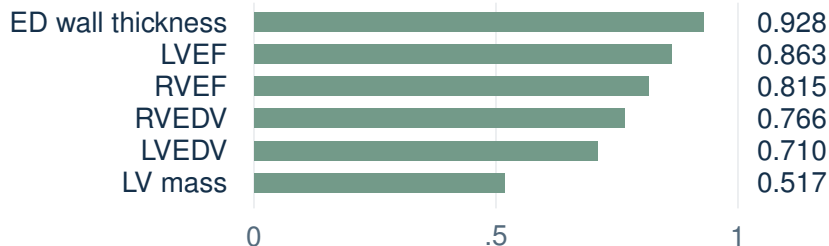


Figure 3. High-impact target correlations (Pearson r , scale 0–1).

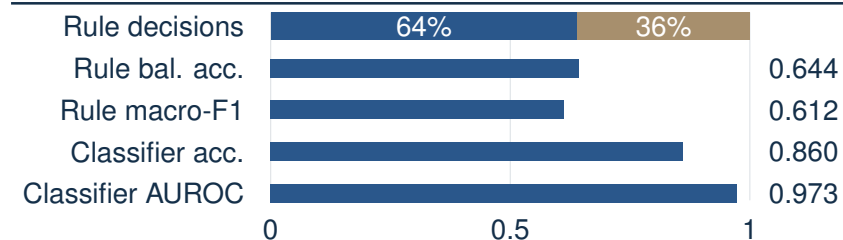
Coverage must accompany rule accuracy

Table 6. Measured-semantic logistic-regression comparator

Split	Accuracy	Bal. acc.	Macro-F1	AUROC	ECE	<i>n</i>
Validation	0.933	0.933	0.931	1.000	0.103	15
Test	0.860	0.860	0.861	0.973	0.179	50

Table 7. 200-rule audit; balanced accuracy and F1 use non-abstained cases

Split	Coverage	Abstain	Bal. acc.	Macro-F1	Conflict	Conf. seen	<i>n</i>
Train	0.565	0.435	1.000	1.000	0.229	0.565	85
Validation	0.667	0.333	1.000	1.000	0.300	0.533	15
Test	0.640	0.360	0.644	0.612	0.313	0.560	50



Held-out uncertainty / 95% CIs

Accuracy: 0.860 [0.76, 0.94]

Balanced accuracy: 0.860 [0.75, 0.95]

Macro-F1: 0.861 [0.75, 0.94]

AUROC: 0.973 [0.947, 0.993]

Row-bootstrap intervals; $n = 50$.

Figure 4. Held-out outcomes: 32 covered; 18 abstained.

AUROC: area under the receiver operating characteristic curve. ECE: expected calibration error (10 equal-mass bins). Conflict: covered cases; conf. seen: all cases during resolution. Perfect selective train/validation scores reflect over-specific granules.

Table 8 with embedded Figure 5.
Physician-predicate evidence

Predicate family	Evidence	Similarity
Abnormal RV / ARVC proxy	Cine + mask	0.9008 
HCM proxy	Cine + mask	0.8564 
MI remodeling proxy	Cine	0.8250 
DCM proxy	Cine + mask	0.7858 
Valvular volume review	Review proxy	0.7825 
Congenital review	Approximation	0.6842 
Conservative normal filter	Research filter	0.5462 
Other families	Unsupported	Unmatched

Bars: admissible similarity on a common 0–1 scale.

Unmatched: restrictive, noncompaction, inflammatory, infiltrative.

HCM / DCM: hypertrophic / dilated cardiomyopathy; MI: myocardial infarction; ARVC: arrhythmogenic right ventricular cardiomyopathy.

Research proxies: support review, not diagnosis or clinician agreement.

Native orientation; no rotation augmentation.

Table 9. Perturbations: 12 cases, 48 runs

Perturbation	Formal change	Semantic change	Mask-count change
Intensity +5%	1.1×10^{-8}	8.8×10^{-6}	0.000000
Shift x , 1 px	0.002591	0.622935	0.000000
Shift y , 1 px	0.002135	0.545662	0.000000
Rotation 2°	0.053114	9.821420	0.000470

$$\Delta_A(\tau) = \frac{1}{|Q|} \sum_{x \in Q} \frac{\|\mathbf{a}(\tau x) - \mathbf{a}(x)\|_2}{\|\mathbf{a}(x)\|_2 + \varepsilon}. \quad (9)$$

$$\Delta_B(\tau) = \frac{1}{28|Q|} \sum_{x \in Q} \|\hat{\mathbf{b}}(\tau x) - \hat{\mathbf{b}}(x)\|_1. \quad (10)$$

Orientation sensitivity

A 2° rotation produces the largest drift despite nearly preserved mask counts.

Q : 12 cases; τ : perturbation; ε : stabilizer. Mixed-unit semantic drift ranks sensitivity only.

What the study establishes

A traceable semantic-to-rule audit chain; strong internal association for selected targets; visible external degradation, conflict, and abstention.

What remains to be validated

Frozen deep encoders; full WEDD; orientation normalization; vendor recalibration; completed expert QC and prospective clinician evaluation.

Current scope

Research auditing only: domain shift and sparse rules limit clinical use.

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Supporting organizations and conference identities

